CLBP-300 Dataset: Technical Specifications and Usage Policy

1. Overview

The **CLBP-300 Dataset** is a high-fidelity scientific resource developed to support research in Remote Blood Pressure (BP) estimation using remote Photoplethysmography (**rPPG**) technology. The data was collected in diverse real-world environments to ensure the highest levels of accuracy and robustness for AI-based physiological monitoring algorithms.

2. Technical Specifications

- **Total Subjects:** 300 participants (244 Males / 56 Females).
- **Total Videos:** 300 high-definition facial recordings.
- **Frame Rate:** 60 Frames Per Second (FPS).
- **Hardware Used:**
    - **iPhone 16 Pro Max** (Mobile recording).
    - **Nikon D5300** (Professional DSLR recording).
- **Ground-Truth Data:** Synchronized measurements for:
    - Systolic Blood Pressure (**SYSBP**).

    - Diastolic Blood Pressure (**DIABP**).
    - Heart Rate (**HR**).
- **Environmental Conditions:** Various indoor and outdoor lighting settings with documented Lux intensity.

3. Data Use Agreement (DUA) - Summary

Access to the dataset is governed by the following legal and ethical terms:

1. **Research Purpose:** The dataset is licensed exclusively for non-commercial, academic research and algorithmic benchmarking.
2. **No Redistribution:** Users are strictly prohibited from sharing, selling, or granting third-party access to the raw video files.
3. **Privacy Protection:** Any attempt to identify participants is strictly forbidden. Public display of raw facial frames or recognizable images in publications is prohibited; only extracted signals (waveforms) may be displayed.
4. **Sustainability Contribution:** The required fee is a contribution to support the ongoing medical research team, equipment maintenance, and clinical data collection cycles.

4. Mandatory Citation

All researchers using this data must cite the following official publication:

Al-Naji, A., et al. "CLBP-300: A Real-World Video Dataset for Cuff-Less Blood Pressure Estimation via rPPG." Preprints 2026. DOI: 10.20944/preprints202604.1079.v1

5. Portal Links

- **Official Portal:** https://sites.google.com/view/clbp-300/home
- **Full Paper:** https://www.preprints.org/manuscript/202604.1079

6. Contact Information: For any inquiries regarding data access, technical support, or potential research collaborations, please contact the Lead Researcher:

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